Here is a summary of the provided GDScript code and an explanation of the game's interface.

📜 Code Summary

The game's logic is split into three main scripts that work together: board.gd (the main controller), move_validator.gd (the rules engine), and victory_checker.gd (the win-condition checker).

1. board.gd (The Main Controller)

This is the most important script, acting as the "brain" of the game. It manages everything the player sees and interacts with.

- **Initialization (_ready):** It builds the visual game board, creates all the UI panels (for scores, captured pieces, etc.), and loads instances of the move_validator and victory_checker scripts .
- **Game State:** It tracks the current_player (white or black) and the current_phase of a turn (Pre-Move Capture, Move, Post-Move Capture) .
- **Turn Flow:** It controls the game's progression. When a player clicks the "End Turn" button, this script advances to the next player , calls the victory_checker to see if anyone won , and then triggers the AI's turn .
- **Player Input (_on_tile_clicked):** This function receives all mouse clicks . It checks the current game phase and routes the click to the correct handler:
    - handle_move_phase_click: Selects a piece and highlights valid moves (in green) .
    - handle_capture_phase_click: Selects a piece and highlights valid captures (in red) .
- **Action Execution:** When a move or capture is attempted, this script first asks the move_validator if it's legal . If yes, it executes the action, moves the pieces, and updates the board state .
- **AI Logic (select_best_move):** This is the AI "brain" . To decide its move, it:
    1. Simulates *every possible legal move* it can make .
    2. For each simulated move, it also simulates all *post-move captures* it could make .
    3. It then scores the final resulting board state using _calculate_position_score , which heavily rewards moves that create winning "fireteam" progressions.
    4. It picks the move that results in the highest score. As a tie-breaker, it prefers moves that "advance" its center of gravity toward the enemy .
- **Undo & Logging:**
    - When a turn begins, start_new_turn_log saves a complete "snapshot" of the entire

board .
    - The **Undo** function (undo_last_turn) works by simply loading that previous snapshot, effectively rewinding time .
    - **Export** (export_game_log_to_text) compiles all logged moves and captures into a readable text file .

2. move_validator.gd (The Rules Engine)

This script is a helper that acts as the game's rulebook. It doesn't manage the game; it just answers "yes" or "no" to questions from board.gd.

- **Move Validation:** The get_valid_moves function checks if a piece's path is the correct distance (e.g., 3 squares for a Square) and that all intermediate squares are empty .
- **Capture Validation:** This is its main job. The get_all_possible_captures function checks for all complex Rithmomachia capture types:
    - **Equality:** Attacker value equals victim value .
    - **Sum/Difference:** Attacker + Helper = Victim, or |Attacker - Helper| = Victim .
    - **Product/Divisor:** Checks if attacker * distance = victim or attacker / distance = victim .
    - **Blockade (Siege):** Checks if the target piece has no legal moves .
- **AI Helper Functions:** It includes functions like get_all_possible_captures_from_state that can check rules on a *simulated* board array, which the AI uses for its planning.

3. victory_checker.gd (The Win Checker)

This is a simple, focused helper script. Its only job is to determine if the game is over.

- **check_for_win:** This is the main function board.gd calls . It checks all win conditions in order.
- **Common Victories:** It checks if the player's captured piece *count* exceeds the "Body" threshold (n0) or if the captured piece *value* exceeds the "Goods" threshold (n1) .
- **Proper Victories:** It checks for the three mathematical progressions (Arithmetical, Geometrical, Harmonic) . To do this, it gets all of the player's pieces currently in enemy territory and checks every 3-piece combination to see if they form a valid sequence.

🖥️ How to Use the Interface

Here is a guide to the layout and functionality available to you as the player.

Before the Game Starts

When you first launch the game, you are presented with a **"Game Setup" window** .

- **Layout:** A simple popup window in the center of the screen.
- **Functionality:** You **must** choose the game's difficulty, which sets the Common Victory conditions .
  - **Short:** Win by capturing 4 pieces (Body) or 100 points (Goods).
  - **Medium (Default):** Win by capturing 10 pieces or 400 points.
  - **Long:** Win by capturing 20 pieces or 1000 points.
- Once you click a button, this window closes, and the game begins.

During the Game

The main game screen is divided into several information panels and a central board.

1. Screen Layout

- **Center:** The 16x8 Rithmomachia game board .
- **Left Panel:** Displays all of **White's** stats:
  - White's score and its breakdown (Captures, Value, Invader Pairs) .
  - A list of all Black pieces White has captured and their total value .
  - The current values remaining in White's Pyramid piece .
- **Right Panel:** Displays all of **Black's** stats (score, captured pieces, pyramid values) .
- **Top-Center Panel:** Shows the current game status:
  - **Victory Condition:** Reminds you of the goal (e.g., "Victory: Medium (Body: 10 pieces, Goods: 400 points)") .
  - **Turn Label:** Shows whose turn it is ("Current Turn: White") .
  - **Action Label:** Tells you what to do now ("Action: pre-move capture") .

- **Top Bar:** Contains the main action buttons:
  - **Phase Button:** A large button whose text changes (e.g., "Done Capturing - Ready to Move" or "End Turn") .
  - **Export Game Log Button:** Saves the game's history .
  - **Undo Last Turn Button:** Reverts the last turn .
- **Top-Right Corner:** A **"?" Help Button** .

2. Game Functionality

Your turn is divided into **three phases**, and your actions are determined by the current phase.

1. **Phase 1: Pre-Move Capture**
   - The **Action Label** shows "Action: pre-move capture."
   - You can click on any of your pieces. If it can capture an enemy piece, the enemy piece will be highlighted in **red** .
   - You can click a red-highlighted enemy piece to capture it. You can do this multiple times if you have several captures available.
   - When you are finished capturing (or if you have no captures), you **click the "Done Capturing - Ready to Move" button** to proceed .
2. **Phase 2: Move**
   - The **Action Label** shows "Action: move." The Phase Button is now disabled .
   - You **must** make one move.
   - Click one of your pieces. All valid destination squares will be highlighted in **green** .
   - Click a green square to move your piece there.
   - As soon as you move, the game *automatically* advances to the next phase.
3. **Phase 3: Post-Move Capture**
   - The **Action Label** shows "Action: post-move capture."
   - This works just like Phase 1. You can again make any captures that your *new* piece positions allow.
   - The **Phase Button** now says "End Turn" . When you are finished, **click "End Turn"** to finalize all your actions and pass the turn to the AI .

3. Other Interface Functions

- **Undo Button:** If you make a mistake (or the AI makes a move you want to take back), clicking **"Undo Last Turn"** will revert the *entire* previous turn . It restores the board from a snapshot taken before the turn began.
- **Export Button:** Clicking **"Export Game Log"** saves a complete text file of every move

and capture from the game to a game_logs folder in the project directory . This is useful for analyzing your game later.

- **Help "?" Button:** Clicking this opens a large **Help Window** with tabs explaining:
  - **Moves:** How each piece type moves .
  - **Captures:** The rules for all capture types .
  - **Victory:** The different ways to win .
  - **Other:** This tab contains special options:
    - **Get Hint Button:** If you're stuck at the *start* of your turn, click this. The AI will analyze the board and show you its recommended move .
    - **Demo Mode Checkbox:** Check this to play as *both* White and Black. This disables the AI so you can play against a friend .
    - **AI vs. AI Checkbox:** Check this to enter "Watch Mode." The AI will play against itself, allowing you to just watch the game .